

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Convert the decimal number (222) into binary. Write your answer without any leading zeros.

1 / 1 point

- ☒ 11011110
- ☐ 1111001
- ☐ 1010101
- ☐ Not possible

✔ Nice work

2. Convert the decimal number 60.75 into binary.

1 / 1 point

- ☐ 110000.11
- ☐ 100100.11
- ☐ 111100.01
- ☒ None of the above are correct

✔ Nice work

3. Perform the base 2 calculation $111101 - 10001$, writing your answer in base 2.

1 / 1 point

- ☐ 11100
- ☒ 101100
- ☐ 01100
- ☐ None of the above are correct

✔ Nice work

4. Perform the base 2 calculation 1110×101 , writing your answer in base 2.

1 / 1 point

- ☐ 1111101
- ☐ 0110001
- ☐ 1000101
- ☒ 1000110

✔ Nice work

5. Convert the decimal number 211.125_{10} to hexadecimal.

1 / 1 point

- ☐ 3D.02
- ☐ D3.02
- ☒ D3.2
- ☐ None of the above are correct

✔ Nice work

We convert the integer part and the fractional part of the number separately.

Integer part:

$211 \div 16 = 13 \text{ rem } 3$

$13 \div 16 = 0 \text{ rem } D$

So, taking the remainders from bottom to top, we have:

$211_{10} = D3$

Fractional part:

$0.125 \times 16 = 2$

So, $0.125_{10} = 0.2_{16}$

Thus,

$211.125_{10} = D3.2_{16}.$